CHM2045 Chapter 2: Atoms and Elements

Law of Conservation of Mass

Definition: In a chemical reaction, matter is neither **created** nor **destroyed**.

$$2\text{Na}(s) + \text{Cl}_2(g) \rightarrow 2\text{NaCl}$$

(reads as "two sodium solids react with one chlorine gas molecule to form two sodium chloride units")

Masses: 7.7 g Na + 11.9 g Cl_2 = 19.6 g NaCl.

Law of Definite Proportions

Definition: A given compound, regardless of source, has the same **proportions** of the constituent atoms.

$$\text{NH}_3 \text{ always has } \frac{14.0 \text{ g N}}{3.0 \text{ g H}} = 4.7$$

(reads as "ammonia always has fourteen point zero grams of nitrogen for every three point zero grams of hydrogen, a ratio of four point seven to one")

This means there are 4.7 grams of nitrogen for every 1 gram of hydrogen in ammonia.

Law of Multiple Proportions

Definition: Atoms combine in small **whole** number ratios.

$$\text{CO}_2$$

(reads as "carbon dioxide")

For carbon dioxide, the ratio is 1 carbon atom to 2 oxygen atoms. You cannot have $\frac{1}{2}$ of an atom (reads as "one-half of an atom").

John Dalton – Atomic Theory (1808)

John Dalton proposed the first modern atomic theory in 1808:

1. Each element is composed of tiny, indestructible particles called **atoms**.
2. All atoms of a given element have a unique **mass** and **properties** that distinguish them from atoms of other elements.
3. Atoms combine in simple **whole** number ratios to form compounds.
4. Atoms from one element **cannot** change into atoms of another element. A chemical reaction only changes the **bonds**, not the identity of the atoms.

J. J. Thomson

In the late 1800s, J. J. Thomson used a cathode ray tube to discover very small, **negatively** charged particles, later called electrons.

Millikan's Oil Drop Experiment (1909)

In 1909, Robert Millikan determined the charge on a single electron and used this value to calculate the mass of an electron.

Rutherford's Gold Foil Experiment (1909)

In 1909, Ernest Rutherford determined the structure of the atom. He proposed that **protons** and **neutrons** are located in a tiny nucleus and that **electrons** are spread out in a cloud around the nucleus. He concluded that:

- Most of the mass of the atom is in the **nucleus**.
- Most of the volume of the atom is **empty space**.
- The atom is neutral; the number of protons equals the number of electrons.

Table 1: Properties of Subatomic Particles

Subatomic Particle	Charge	Mass (g)	Mass (amu)
Proton	+1	1.67262×10^{-24} (reads as "one point six seven two six two times ten to the negative twenty-four")	1.00727
Neutron	0	1.67493×10^{-24} (reads as "one point six seven four nine three times ten to the negative twenty-four")	1.00866
Electron	-1	9.1×10^{-28} (reads as "nine point one times ten to the negative twenty-eight")	0.00055

Elements

Defined by their number of *protons*.

Atomic number (Z) — the number of protons in an atom. The atomic number appears above the element symbol on the periodic table. Elements are listed in the periodic table in order of increasing atomic number.

Elements are represented by one or two letters. The first letter is capitalized.

Isotopes

The number of *neutrons* can vary among atoms of the same element.

Isotopes occur naturally for all elements.

Representing Isotopes

Isotopes can be represented in several ways:

- **Mass number (A)** = number of protons plus number of neutrons.

  $$A = p + n$$

  (reads as "A equals p plus n")
- **Element symbol (X)** represents the chemical symbol of the element.
- **Atomic number (Z)** = number of protons.

  $$Z = p$$

  (reads as "Z equals p")
- **Number of electrons in a neutral atom** equals the number of protons.

  $$e^- = p$$

  (reads as "e minus equals p")

Example (Neon-20):

- Atomic number:

  $$Z = 10$$

  (reads as "Z equals ten")
- Protons:

  $$p = 10$$

  (reads as "p equals ten")
- Mass number:

  $$A = 20$$

  (reads as "A equals twenty")
- Neutrons:

  $$n = 10$$

  (reads as "n equals ten")

Other common notations:

- Neon-20
- Ne-20

Isotopes usually have the *same* chemical behavior but *different* natural abundances.

Example 1: Argon-40

(a) How many neutrons does Ar-40 have?

Atomic number of argon:

$$Z = 18$$

(reads as "Z equals eighteen")

Neutrons:

$$40 - 18 = 22$$

(reads as "forty minus eighteen equals twenty-two")

So Ar-40 has 22 neutrons.

(b) If Ar is neutral, how many electrons does this isotope have?

$$e^- = 18$$

(reads as "e minus equals eighteen")

Example 2: Chlorine Isotope with 18 Neutrons

What are the atomic number (Z), mass number (A), and symbol?

Chlorine has atomic number:

$$Z = 17$$

(reads as "Z equals seventeen")

Mass number:

$$17 + 18 = 35$$

(reads as "seventeen plus eighteen equals thirty-five")

Symbol:

$$^{35}_{17}\mathrm{Cl}$$

(reads as "chlorine-thirty-five with atomic number seventeen")

Ions

The number of *electrons* can vary for atoms of the same element. Atoms can gain or lose electrons to form ions.

Table 1: Comparison of Cations and Anions

Property	Cations	Anions
Electron change	Lose electrons	Gain electrons
Charge	Positive charge	Negative charge
Commonly formed by	Metals	Nonmetals

Example: Oxide Ion (O^2)

How many protons and electrons are present in O^2 (oxygen two minus)?

Atomic number of oxygen:

$$Z = 8$$

(reads as "Z equals eight")

Number of protons:

$$p = 8$$

(reads as "p equals eight")

Because the ion has a $2-$ charge, it has gained two electrons:

$$e^- = 10$$

(reads as "e minus equals ten")

Development of the Periodic Table

Dmitri Mendeleev in the 1800s organized the periodic table based on approximately 65 elements known at that time. He arranged them according to increasing atomic mass and similar chemical properties.

Note the following features on a periodic table:

- Period
- Group (family)
- Metals
- Transition metals
- Metalloids
- Non-metals
- Inner transition metals

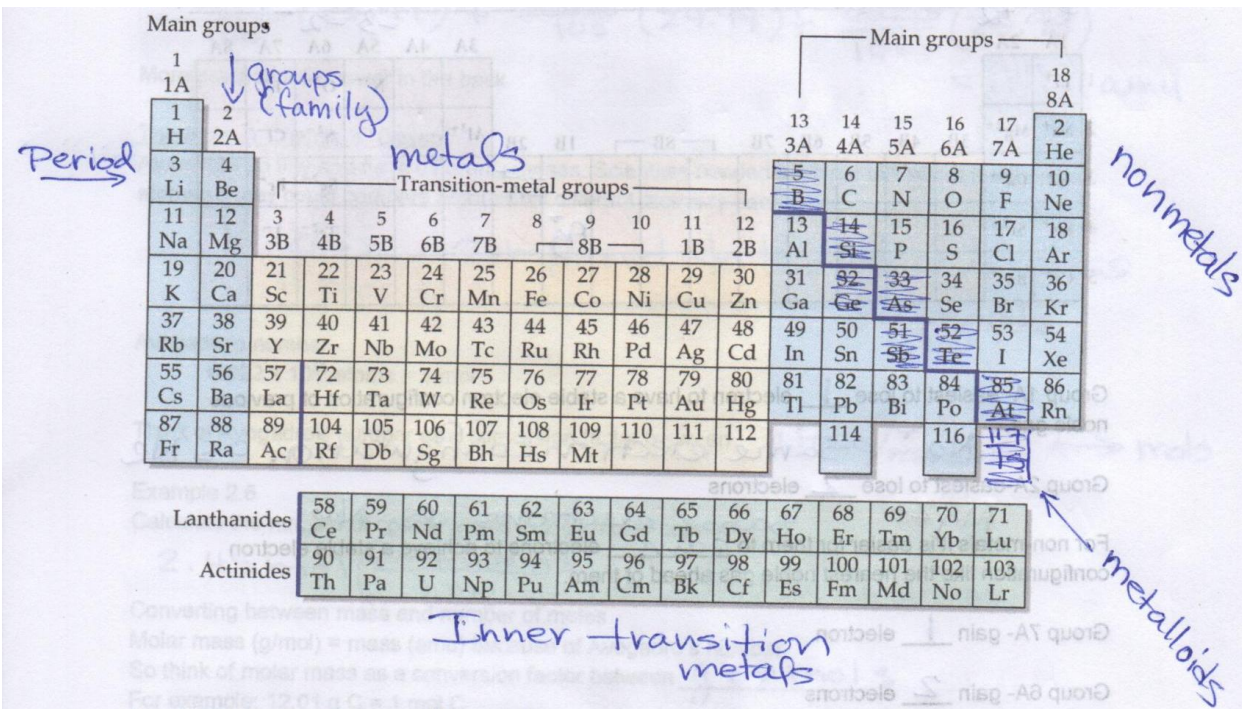

Figure 1: Periodic table highlighting main groups (1A–8A), transition metal groups (3B–2B), and inner transition metals (lanthanides and actinides). Handwritten annotations label groups (families), periods, metals, nonmetals, metalloids, and inner transition metals.

- **1A – Alkali Metals:** React violently with H O
- **2A – Alkaline Earth Metals**
- **7A – Halogens**
- **8A – Noble Gases:** Unreactive; stable

Ions and the Periodic Table

Because noble gases have a stable electron configuration, other atoms try to gain or lose electrons to be like the nearest noble gas.

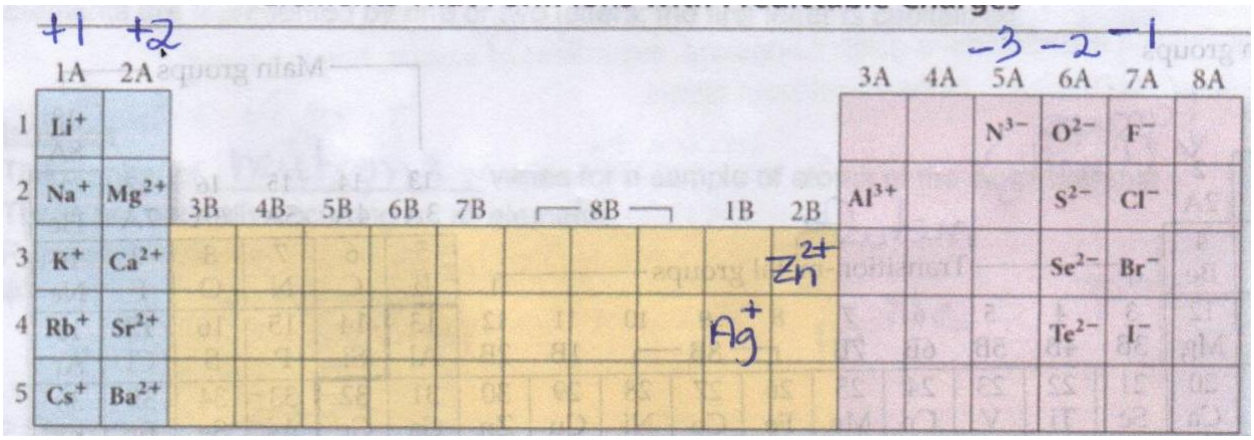

Figure 2: Simplified periodic table showing elements that form ions with predictable charges. Group 1A elements (Li , Na , K , Rb , Cs) form +1 ions; Group 2A elements (Mg² , Ca² , Sr² , Ba²) form +2 ions; Group 3A includes Al³ ; Group 5A elements form 3− ions (such as N³); Group 6A form 2− ions (such as O² , S² , Se² , Te²); and Group 7A form 1− ions (such as F , Cl , Br , I). Transition metals such as Zn² and Ag are also indicated.

Elements That Form Ions with Predictable Charges

- **Group 1A:** Easiest to lose 1 electron to have a stable electron configuration of the previous noble gas. For example, Na has the same electron configuration as Ne.
- **Group 2A:** Easiest to lose 2 electrons.
- **Nonmetals:** It is easier for them to gain electrons to achieve a stable electron configuration like the nearest noble gas ahead of them.
- **Group 7A:** Gain 1 electron.
- **Group 6A:** Gain 2 electrons.
- **Group 5A:** Gain 3 electrons.

Remember that N^{3-} still has 7 protons, but when it forms a stable ion it has 10 electrons (reads as "N three minus").

Calculating Average Atomic Mass

This is a weighted average of the naturally occurring isotopes for an element; it is the number shown under the symbol on the periodic table.

$$\text{Average atomic mass} = \frac{(\%\text{ abundance isotope 1}) \times (\text{mass isotope 1})}{100} + \frac{(\%\text{ abundance isotope 2}) \times (\text{mass isotope 2})}{100}$$

(reads as "average atomic mass equals the percent abundance of isotope one times the mass of isotope one divided by one hundred, plus the percent abundance of isotope two times the mass of isotope two divided by one hundred")

Practice 2.5

Magnesium has three naturally occurring isotopes with the following masses and abundances:

Isotope Mass (amu)	Abundance (%)
23.99	78.99%
24.99	10.00%
25.98	11.01%

Calculate the average atomic mass of magnesium.

Solution:

$$\frac{78.99}{100}(23.99) + \frac{10.00}{100}(24.99) + \frac{11.01}{100}(25.98) = 24.31 \text{ amu}$$

(reads as "seventy-eight point nine nine divided by one hundred times twenty-three point nine nine, plus ten point zero zero divided by one hundred times twenty-four point nine nine, plus eleven point zero one divided by one hundred times twenty-five point nine eight equals twenty-four point three one atomic mass units")

The weighted average is calculated by multiplying each isotope's mass by its decimal abundance and summing the results, giving an average atomic mass of 24.31 amu (atomic mass units).

Additional Practice 2.5 answers are available in the answer key section at the back of the book.

The Mole: A Chemist's "Dozen"

Atoms are extremely small and have different masses. Scientists needed a way to describe large numbers of atoms so they could compare amounts of different elements quantitatively.

For example, saying "one-half of the class compared to the other half of the class" illustrates comparing equal portions of a group.

Avogadro's Number

Avogadro's number is:

$$6.022 \times 10^{23} \text{ atoms} = 1 \text{ mol}$$

(reads as "six point zero two two times ten to the twenty-third atoms equals one mole")

A mole (mol) is a counting unit used in chemistry. Avogadro's number, 6.022×10^{23}, represents the number of atoms, molecules, or other particles in one mole. It can be used as a conversion factor between atoms or molecules and moles.

Example 2.6

Calculate the number of copper (Cu) atoms in 2.45 mol of copper.

Solution:

$$2.45 \text{ mol} \left(\frac{6.022 \times 10^{23} \text{ atoms}}{1 \text{ mol}}\right) = 1.48 \times 10^{24} \text{ Cu atoms}$$

(reads as "two point four five moles times six point zero two two times ten to the twenty-third atoms per one mole equals one point four eight times ten to the twenty-fourth copper atoms")

The moles cancel, leaving the total number of copper atoms.

Converting Between Mass and Number of Moles

Molar mass (in grams per mole, g/mol) is numerically equal to the atomic mass (in atomic mass units, amu) because of Avogadro's number.

Molar mass can be used as a conversion factor between grams and moles.

For example:

12.01 g C = 1 mol C

(reads as "twelve point zero one grams of carbon equals one mole of carbon")

Practice 2.7

Calculate the amount of copper in moles in a 35.8 g pure copper sheet.

Solution:

$$35.8 \text{ g Cu} \left(\frac{1 \text{ mol Cu}}{63.546 \text{ g}}\right) = 0.563 \text{ mol Cu}$$

(reads as "thirty-five point eight grams of copper times one mole of copper per sixty-three point five four six grams equals zero point five six three moles of copper")

Practice 2.8

How many carbon atoms are there in a 1.3 carat diamond? (1 carat = 0.20 g)

Solution:

$$1.3 \text{ carats} \left(\frac{0.20 \text{ g}}{1 \text{ carat}}\right) \left(\frac{1 \text{ mol C}}{12.011 \text{ g}}\right) \left(\frac{6.022 \times 10^{23} \text{ atoms}}{1 \text{ mol}}\right) = 1.3 \times 10^{22} \text{ atoms}$$

(reads as "one point three carats times zero point two zero grams per carat times one mole of carbon per twelve point zero one one grams times six point zero two two times ten to the twenty-third atoms per mole equals one point three times ten to the twenty-second atoms")

Periodic Table